NON-CONVENTIONAL ENERGY SOURCES

Subject Code: 23A03703a | UNIT – V : OCEAN ENERGY & DIRECT ENERGY CONVERSION

1. Ocean Energy – Overview

- Oceans cover 71% of earth and act as a huge solar collector & store. Forms: thermal (OTEC), tidal, waves, currents, salinity gradient.
- India: coastline 7500 km; OTEC ≈ 180,000 MW; tidal ≈ 8,000–9,000 MW (Gulf of Khambhat, Gulf of Kutch, Sundarbans); wave ≈ 40,000 MW.

2. OTEC – Ocean Thermal Energy Conversion

> OTEC uses the **temperature difference** between warm surface water (25–30 °C) and cold deep water (4–6 °C at 600–1000 m depth) in tropical oceans to run a heat engine.

❖Principle & utilisation

- $\Delta T \geq 20$ °C is the minimum needed – available between ± 20° latitude (India: Lakshadweep, Andaman, Tamil Nadu coast).
- Max (Carnot) efficiency: $\eta = 1 - T_c/T_h = 1 - 278/298 = 6.7$ % (actual net 2–3 %) – very low, so large water flows needed (~ 4 m^3/s per MW).
- **By-products:** desalinated water, air-conditioning, mariculture (nutrient-rich deep water), hydrogen/ ammonia.
- Power consumed by pumps (cold-water pipe 1000 m, warm-water pump) is 20–30% of gross output.

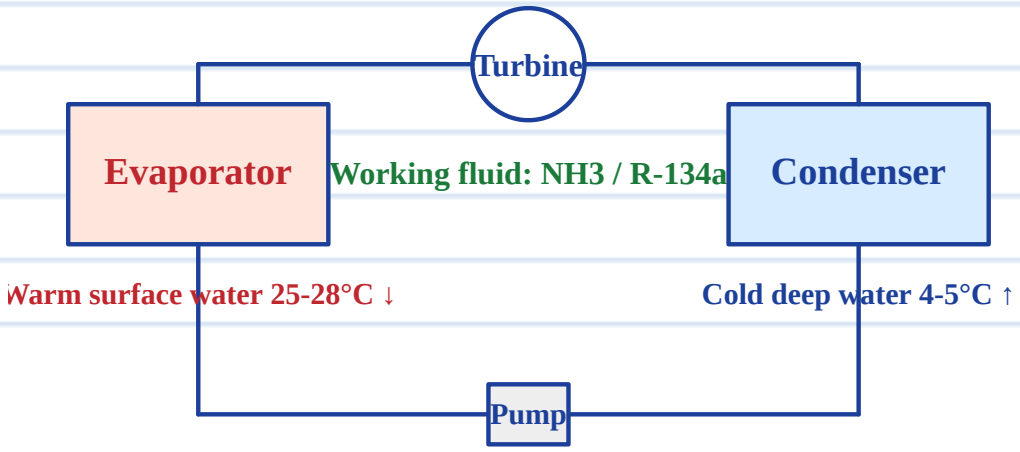
❖Setting of OTEC plants (site/ location)

- **Land-based:** shore plant near deep water; long cold-water pipe along the sea bed; stable, easy transmission, but pipe cost.
- **Shelf-mounted:** on continental shelf, fixed platform.
- **Floating (offshore):** barge/ spar/ ship anchored in deep water (e.g. India's 1 MW floating plant off Tamil Nadu 2001, Kavaratti 65 kW).
- Needs: $\Delta T > 20$ °C within short distance (< 10 km) of coast, deep water close to shore, stable sea state, near load.

- **Materials:** titanium/ aluminium heat exchangers (anti-corrosion, bio-fouling control – chlorination), HDPE/ FRP cold water pipe (dia 10 m for 100 MW).

3. OTEC Thermodynamic Cycles

❖(a) Closed cycle (Anderson cycle) – Rankine



Closed (Rankine) cycle

Fig 5.1 Closed-cycle OTEC plant

- Working fluid with low boiling point (**ammonia, propane, R-134a**) circulates in a closed loop:

1. **Evaporator:** warm sea water boils the fluid;
2. high-pressure vapour expands in **turbine** → **generator**;
3. **Condenser:** cold deep sea water condenses vapour;
4. **Pump** returns liquid to evaporator.

- Compact turbine, no direct sea water contact; heat exchangers are large.
- Net work: $W_{net} = W_{turbine} - W_{pump} - W_{sea\text{-}water\ pumps}$

❖(b) Open cycle (Claude cycle)

- **Warm sea water itself** is the working fluid. It is **flash-evaporated** in a vacuum chamber (~ 2.4 kPa, boiling at ~ 21 °C); the low-pressure steam drives a **huge low-pressure turbine**, then is condensed by cold water (direct-contact or surface) → **fresh water** produced as by-product.
- Advantages: desalinated water, no heat exchanger fouling. Disadvantages: very large turbine (low pressure), vacuum maintenance, deaeration.

❖(c) Hybrid cycle

- Open flash evaporator for steam + closed-cycle ammonia loop to generate power; gives both **power & fresh water** with a smaller turbine.

❖Comparison

Feature	Closed	Open
Working fluid	NH_3/ propane	Sea water steam
Turbine size	Small	Very large
Fresh water	No	Yes
Efficiency	2–3%	1–2%
Pressure	High	Vacuum

4. Tidal Energy

> Tides are periodic rise and fall of sea level caused by the **gravitational pull of moon (and sun)** on oceans together with earth's rotation. Two high & two low tides per ~ 24 h 50 min.

- **Spring tide:** sun, moon, earth aligned (new/ full moon) → high range. **Neap tide:** perpendicular (quarter moon) → low range.
- Tidal range for power: **> 5 m**; suitable sites: Bay of Fundy (Canada, 16 m), La Rance (France, 240 MW, since 1966), Kislaya Guba, Sihwa (Korea 254 MW), **Gulf of Khambhat (7–11 m)**, Kutch (5–8 m), Sundarbans.

❖Power from a tidal basin

Energy per tide $E = \frac{1}{2}\,\rho\, g\, A\, R^2$ (A = basin area, R = tidal range)

- Average power $P = \rho\, g\, A\, R^2 / (2\,T)$, T = tide period (12 h 25 min = 44,700 s). With efficiency η → $P = \eta \times$ above.
- Example: $A = 10\ km^2$, $R = 6$ m, $\rho = 1025$: $E = \frac{1}{2} \times 1025 \times 9.81 \times 10^7 \times 36 = 1.81 \times 10^{12}$ J/tide; $P_{avg} = 1.81 \times 10^{12} / 44700 = 40.5$ **MW** (ideal).

❖Tidal power plant types

- **Single-basin, single-effect:** one basin, one gate; generate on ebb only (or flood only) – 1 or 2 times/day, intermittent.
- **Single-basin double-effect:** generate on both ebb & flood with bulb/ reversible turbines.

- **Double-basin:** high basin + low basin with turbines in between → continuous (firm) power; costly.
- **Tidal stream/ barrage-free turbines** (marine current turbines), **tidal lagoons** (Swansea).
- **Components:** barrage/ dam, sluice gates, **bulb turbines (Kaplan type)**, generator, transformer, ship locks.

❖Advantages / Disadvantages

- Predictable, no fuel, long life, no pollution. High capital cost, intermittent, limited sites, harms marine life/ silting, corrosion of equipment.

5. Wave Energy

> Waves are created by **wind blowing over sea surface**; wave energy is concentrated form of wind (solar) energy. Typical power in deep water: $P = \rho\, g^2\, H^2\, T / (32\pi) \approx 0.5\, H^2\, T$ **kW per metre** of wave front (H = wave height m, T = period s).

- Example: $H = 2$ m, $T = 8$ s → $P = 0.5 \times 4 \times 8 =$ **16 kW/m**. Good sites 20–70 kW/m (Atlantic); India west coast ≈ 5–15 kW/m (Vizhinjam 150 kW OWC plant).

❖Wave energy conversion devices

- **Oscillating water column (OWC):** waves force air in a chamber through a **Wells turbine** (rotates same way for both air directions) → generator. (Shore/ breakwater – Vizhinjam, Pico, LIMPET).
- **Salter's duck (nodding duck):** cam-shaped float that nods with wave, converts rocking into rotation – ~ 90% efficiency in lab.
- **Cockerell raft:** hinged floating rafts; relative motion drives hydraulic pumps.
- **Pelamis (snake):** articulated cylinders; hydraulic rams – generators.
- **Point absorber / heaving buoy (Archimedes wave swing, Power buoy), wave-dragon (overtopping device):** waves overtop a ramp into a reservoir, flow back through low-head turbines.
- **Wave pump, Dam-Atoll, Pressure differential devices.**
- **Problems:** irregular waves, survivability in storms, corrosion, transmission, high cost.

6. Mini-Hydel Power Plants

Hydro plants of small capacity: **Micro** < 100 kW; **Mini** 100 kW – 2 MW (India: up to 3 MW); **Small** 2 – 25 MW. Use **run-of-river** or canal-fall/ small dam schemes. India potential ≈ 21,000 MW (SHP), ~ 5 GW installed.

$P = \eta \rho g Q H$ [W] (ρ = 1000 kg/m^3, Q = discharge m^3/s, H = net head m) $\approx 9.81 \eta Q H$ kW

- **Components:** diversion weir/ intake, **desilting tank**, canal/ **penstock**, forebay, **turbine** (Pelton – high head > 100 m; Francis – medium 30–200; Kaplan/ propeller/ cross-flow – low head < 30 m), generator, tailrace, controller (ELC).
- **Types:** low head (< 15 m), medium head (15–70 m), high head (> 70 m).

❖Economics

- Capital cost ≈ ₹ 6–10 crore/MW; low O & M, life 35–50 years, CUF 40–50%, **cost per kWh** ₹ 2–4; no fuel; gestation 2–3 years.
- Benefits: rural electrification, irrigation, flood/ no submergence, local employment; weak sides: seasonal flow, silt, site-specific, environmental flow need.

❖Example

Q = 2 m^3/s, H = 30 m, η = 0.8: $P = 9.81 \times 0.8 \times 2 \times 30$ = 471 kW; annual energy at CUF 0.5 = $471 \times 8760 \times 0.5 = 2.06 \times 10^6$ kWh.

7. DIRECT ENERGY CONVERSION (DEC) – Need & Principles

DEC = converting one form of energy **directly into electricity** without an intermediate mechanical (rotating) stage – no boiler/ turbine/ generator.

❖Need for DEC

- Conventional thermal plant: heat → mechanical → electrical; limited by **Carnot efficiency** and has many losses ($\eta \approx$ 30–40%), moving parts, pollution.
- DEC: **fewer conversion steps, no moving parts, silent, compact, reliable, low maintenance**, can reach higher efficiency (fuel cell 40–60%), useful for **space, remote & military** power.

- Exploits sources like heat, light, chemical energy, plasma.

❖Carnot cycle & its limitations

- Carnot efficiency: $\eta_c = 1 - T_L / T_H$ – maximum for any heat engine between T_H and T_L.
- Limitations in practice: need **isothermal heat transfer** (infinite area/ time), reversible processes, **no real working fluid** can follow it, low work ratio, material limits on T_H (turbine blade ~ 600–1000 °C), T_L limited by ambient (~ 300 K).
- **Fuel cells & PV** convert chemical/ light energy directly and are **not Carnot-limited** (limit set by $\Delta G/\Delta H$ or band gap).

❖Types of DEC

Device	Conversion
Thermoelectric generator	Heat → electricity (Seebeck)
Thermionic generator	Heat → electrons emitted
MHD generator	Heat/ kinetic of plasma → electricity
Fuel cell	Chemical → electricity
Photovoltaic	Light → electricity
Piezo/ Electrogasdynamic (EGD)	Mechanical/ gas flow → electricity

8. Thermo-electric Generators (TEG)

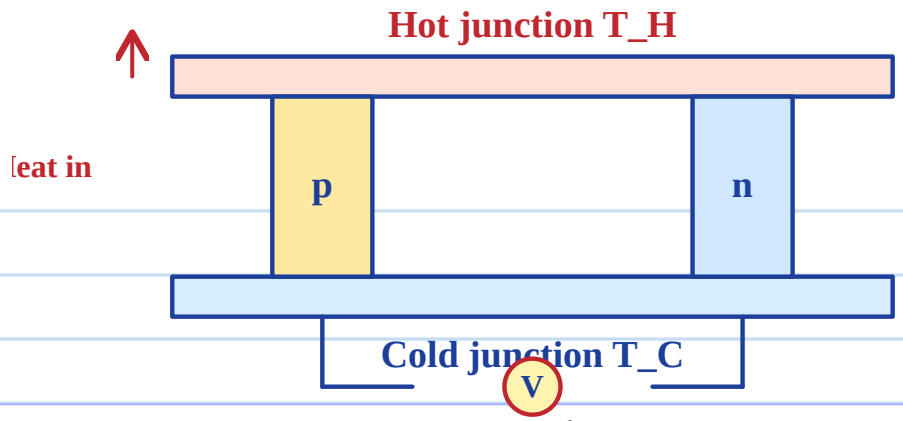


Fig 5.2 Thermoelectric generator (p-n couple, Seebeck effect)

❖Three thermo-electric effects

- **Seebeck effect (1821):** if two dissimilar conductors/ semiconductors are joined and the two junctions are kept at **different temperatures**, an **emf** is produced. $V = \alpha_{AB} (T_H - T_C)$, α = Seebeck coefficient ($\mu V/K$).

- Peltier effect (1834): reverse – when current passes through a junction of two dissimilar conductors, heat is absorbed or evolved at the junction (used in thermoelectric coolers). $Q = \Pi\, I = \alpha\, T\, I$ (Π = Peltier coefficient).
- Thomson effect (1851): heat absorbed/ released when current flows through a single conductor having a temperature gradient. $Q = \tau\, I\, dT/dx$.
- Kelvin relations: $\Pi = \alpha\, T$; $\tau = T\, d\alpha/dT$.
- Joule heating I^2R is irreversible and always present. (Note: "Joule–Thomson effect" in your syllabus refers to the cooling/ heating of a real gas when it expands through a throttle at constant enthalpy: $\mu_{JT} = (\partial T/\partial P)_h$; cooling when $\mu > 0$ (below inversion temp) – basis of liquefaction & refrigeration; $\Delta T \approx \mu\, \Delta P$.)

❖TEG operation

- Many p-type & n-type semiconductor legs connected electrically in series, thermally in parallel between hot plate (T_H) and cold plate (T_C). Heat flows from hot to cold side; carriers diffuse to cold end → voltage ≈ 0.2–0.4 mV/K per couple.

❖Efficiency & Figure of merit

Figure of merit $Z = \alpha^2 / (\rho\, k) = \alpha^2\, \sigma / k$ [1/K] ; ZT dimensionless

- α = Seebeck coeff; σ = electrical conductivity (ρ = resistivity); k = thermal conductivity.
- Maximum efficiency:

$$\eta_{max} = (T_H - T_C)/T_H \times [\sqrt{(1+ZT)} - 1]/[\sqrt{(1+ZT)} + T_C/T_H],\ T^- = (T_H+T_C)/2$$

- Larger ZT ⇒ closer to Carnot. Good materials have high σ, high α, low k ("phonon-glass electron-crystal"). $ZT \approx 1$ now (needs > 3 to compete with engines).
- Materials: Bi_2Te_3 (up to 200 °C, $ZT \approx 1$), PbTe (300–600 °C), SiGe (1000 °C – space RTGs), skutterudites, half-Heusler, TAGS, oxides.
- Power: $P = I^2 R_L$ with maximum when load R_L = internal R.

❖Applications

- Radioisotope thermoelectric generators (RTG) in Voyager, Cassini, Mars rovers; waste-heat recovery in automobiles & industry; remote pipeline/ telecom power; wearable/ body-heat devices; cooling (Peltier) – mini-fridge, CPU cooler, wine coolers; gas stove generators.
- Efficiency 5–8% – low but reliable.

9. Thermionic Converter (brief)

- A hot emitter (1500–2000 K) emits electrons by thermionic emission (Richardson: $J = A T^2 \exp(-W/kT)$) across a small gap to a cooler collector; space-charge neutralised with caesium vapour. $\eta \approx 10–20\%$.

10. MHD Generators

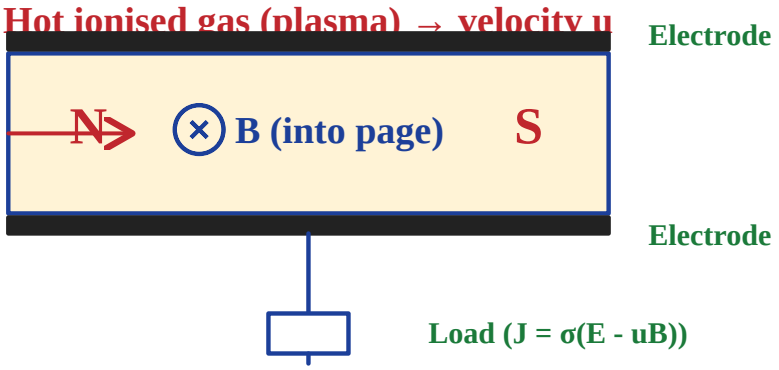


Fig 5.3 MHD generator (Faraday type) – flow, field and current mutually ⊥

> Magneto-hydrodynamic (MHD) generator converts the kinetic/ thermal energy of a hot, electrically conducting fluid (plasma) directly into electricity by passing it across a magnetic field – Faraday's law of induction applied to a moving fluid conductor instead of a copper coil.

❖Principle

- A conductor moving with velocity u perpendicular to field B gets an induced emf $E = u B$ (per unit length), acting perpendicular to both (Fleming's right-hand rule); electrodes on the channel walls collect the current → load.
- Current density: $J = \sigma (u B - E_y)$; open-circuit voltage $V_{oc} = u B d$ (d = electrode gap).
- Electrical power per unit volume = $\sigma u^2 B^2 K (1 - K)$ where load factor $K = E/(uB)$; max power at $K = 0.5$.

P_{max} per volume = $\sigma u^2 B^2 / 4$

❖Dissociation and ionisation

- At > 2500–3000 K, gas molecules **dissociate** (e.g. $O_2 \rightarrow 2O$) and then **ionise** (atoms → ions + e^-) – producing a **plasma** with free electrons (conductive).
- Combustion gases at 2500 K have very low conductivity, so the gas is **seeded** with a **low ionisation potential** alkali metal (**potassium or caesium carbonate, ~ 1% by weight**) → $\sigma \approx 10$–50 S/m.
- **Saha equation** gives degree of ionisation: $n_e\, n_i / n_0 = (2\pi m_e kT/h^2)^{3/2} \exp(-E_i/kT)$. Hence plasma needs high T, low E_i, seed.
- Non-equilibrium plasma (high electron temp) can give high σ at lower gas temp.

❖Hall effect

- In a strongly ionised gas the current J itself interacts with B: a **Hall field** E_x develops along the flow direction (**Hall effect**), because the **Lorentz force** $v \times B$ deflects the charge carriers. Hall parameter $\beta = \omega_c \tau = e B \tau / m_e$ (cyclotron frequency × collision time).
- Hall current short-circuits the axial field in a continuous-electrode (Faraday) channel, reducing output → use **segmented electrodes** (Faraday), **Hall generator** (Hall current collected by electrodes at ends), or **disc generator**.

❖Types of MHD generator

- **Faraday** (continuous or segmented electrode), **Hall**, and **Disc** (Corbino) types.
- **Open cycle:** combustion products (coal/ oil + seed) → nozzle → MHD channel (2500–3000 K) → air preheater/ steam bottoming plant → seed recovery; the gas is discharged.
- **Closed cycle:** inert gas (argon/ helium + Cs seed) circulates (T ≈ 1800–2000 K); heated in a **heat exchanger** by nuclear or coal; **non-equilibrium ionisation**; no pollution; seed not lost. Also **liquid-metal MHD** (two-phase).

❖MHD power-generation system

- **Components:** combustor/ heat source, **seeding**, **nozzle** (accelerate to ~ 1000 m/s, Mach 1–2), **MHD channel** inside superconducting/ electromagnet (B = 4–6 T), **inverter** (DC → AC), **diffuser**, air preheater, **steam bottoming cycle**, **seed recovery** unit, compressor.
- Combined MHD + steam plant efficiency **50–60 %** (MHD topping ~ 20–25%).
- **Magnetic flux** ($\Phi = B\,A$ in Weber) must be high and uniform in the channel; stronger B ⇒ more power since $P \propto B^2$.
- **Magnetic flux density** is produced by **superconducting magnet** (NbTi at 4 K) to avoid copper loss.

❖MHD accelerator & MHD engine

- If the **current is fed in** from an external source (J × B force), the **Lorentz force accelerates** the fluid: $F = J \times B$ – the **MHD accelerator / pump** (used for liquid-metal pumping in nuclear reactors, **MHD propulsion** of ships, plasma thrusters in space = **MHD engine**).
- Thrust $F = B\,I\,d$ (I = current, d = electrode gap). The machine is the **reverse of the generator** (motor action).

❖Merits & problems

- **Merits:** high efficiency (topping), no moving parts at high T, quick start-up, less pollution (SO_x locked in seed), saves water cooling, can use coal.
- **Problems:** very high temperature materials (ceramics, ZrO_2/ MgO electrodes erosion/ corrosion), strong magnets, seed recovery, nozzle erosion, low σ, voltage leakage/ arcing, high initial cost.

❖Economic aspects

- Capital ₹ is higher for a pilot MHD plant but **fuel saving 10–15%** and lower emissions give lower cost per kWh at scale; research units in India (BHEL Tiruchirappalli 5 MW pilot), Russia U-25 (25 MW), USA.

11. Electro-Gas-Dynamic (EGD) Conversion

- Charged particles (ions) are injected into a high-velocity gas stream and carried against an electric field by the flow; the gas does work on the charges → DC at high voltage (10–100 kV), tiny current. No magnetic field needed. Efficiency 5–10%. Uses: Van de Graaff-like generators, electrostatic precipitators & filtering.

12. Fuel Cells

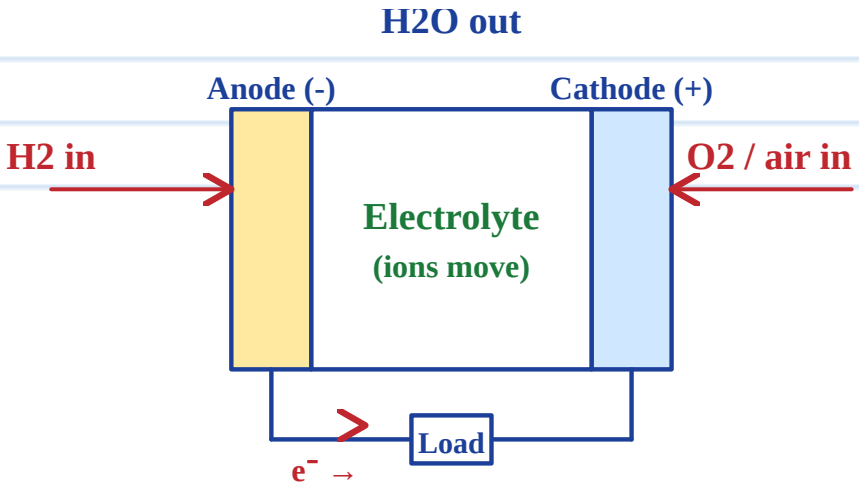


Fig 5.4 Hydrogen-oxygen fuel cell

> A fuel cell is an electrochemical device that converts the chemical energy of a fuel (H_2, CH_4, methanol) and oxidant (O_2/ air) directly into electrical energy, continuously, as long as reactants are supplied (unlike a battery which stores a limited amount).

❖Principle (H_2–O_2 acid-electrolyte cell)

- Anode (–): $H_2 \rightarrow 2H^+ + 2e^-$
- Cathode (+): $\frac{1}{2}O_2 + 2H^+ + 2e^- \rightarrow H_2O$
- Overall: $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ + electrical energy + heat
- Electrons travel through the external circuit (load); ions through the electrolyte; catalyst (Pt) on porous electrodes speeds up reactions.

❖Faraday's laws & thermodynamic aspects

- Faraday's first law: mass of substance liberated ∝ charge passed. Second law: mass ∝ equivalent weight.

$$m = ZIt = (M/nF)\cdot It\ ;\ F = 96{,}485\ C/mol\ e^-$$

- For H_2: $n = 2$ electrons per molecule → consumption = $I/(2F)$ mol/s per cell = ≈ 1.04×10^{-8} kg/s per amp.
- Reversible (open-circuit) cell voltage: $E = -\Delta G / (n F)$. For H_2–O_2 at 25 °C: $\Delta G = -237.2$ kJ/mol → $E = 237200/(2 \times 96485) = 1.23$ V (liquid water; $\Delta H = -285.8$ kJ/mol → thermoneutral voltage 1.48 V).
- Maximum (thermodynamic) efficiency:

$$\eta_{max} = \Delta G / \Delta H = 237.2 / 285.8 = 83\ \% \text{ (at 25 °C; falls with T)}$$

- Actual efficiency $\eta = \eta_{thermo} \times (V_{actual}/E) \times$ fuel utilisation ≈ 40–60% (up to 85% with CHP).
- Not limited by Carnot since no heat engine.
- Voltage losses (polarisation): (i) activation (sluggish reaction, at low current), (ii) ohmic (electrolyte/ electrode resistance, mid current), (iii) concentration (mass transport, at high current). Operating $V \approx 0.6$–0.8 V per cell.
- Power $P = V I$; many cells stacked in series (bipolar plates) to get useful voltage.
- Effect of T & P: higher T lowers E (−0.85 mV/K) but speeds kinetics; higher P raises E: $\Delta E = (RT/2F) \ln(P_{H2} P_{O2}^{1/2}/P_{H2O})$ (Nernst).

❖Selection of fuels

- Hydrogen – best (high reactivity, clean) – by electrolysis, steam reforming, storage as compressed gas/ metal hydrides.
- Methanol (DMFC), natural gas/ methane/ hydrocarbons via internal/ external reforming, ammonia, hydrazine, biogas, coal gas (for high-T cells).
- Requirements: high reactivity at anode, availability, low cost, easy storage, no electrode poisoning (CO poisons Pt in PEMFC, S compounds poison catalysts) → purify.

❖Operating conditions & types

Type	Electrolyte	Temp	Fuel	Use
Alkaline (AFC)	KOH	60–90 °C	pure H_2	Apollo, space
PEMFC	Polymer (Nafion)	60–90 °C	H_2	Cars, portable
PAFC	Phosphoric acid	150–220 °C	H_2/ reformate	CHP buildings (200 kW)
MCFC	Molten carbonate	~ 650 °C	CH_4/ coal gas	Power plants
SOFC	Zirconia ceramic (YSZ)	800–1000 °C	CH_4, CO, H_2	Stationary, high efficiency
DMFC	Polymer	60–120 °C	Methanol	Small portable

- **Advantages:** high efficiency, silent, clean (H_2O only), modular, no moving parts, fast load response, CHP.
- **Disadvantages:** high cost (Pt, membranes), H_2 storage/ infrastructure, catalyst poisoning, durability, water management.
- **Applications:** spacecraft power & drinking water, buses/ cars (Toyota Mirai), backup & distributed power, submarines, laptops/ mobile.

❖Example

A cell gives 0.7 V at 50 A. P = 35 W per cell. For 5 kW need 5000/35 = 143 cells in series. H_2 consumption per cell = $50 \times 1.04\times10^{-8} = 5.2\times10^{-7}$ kg/s → 0.0019 kg/h per cell. Efficiency (vs. HHV 1.48 V) = 0.7/1.48 = 47 %.

13. Solved Problems

❖Problem 1 – OTEC Carnot

T_w = 27 °C (300 K), T_c = 5 °C (278 K): η_c = 22/300 = 7.3 %.

❖Problem 2 – TEG figure of merit

α = 200 μV/K, ρ = 1×10^{-5} Ω·m, k = 1.5 W/mK: $Z = (2\times10^{-4})^2/(10^{-5}\times1.5) = 4\times10^{-8}/1.5\times10^{-5} = 2.67\times10^{-3}$ K^{-1}; at T = 400 K → ZT = 1.07.

❖Problem 3 – Seebeck voltage

100 couples each α = 400 μV/K, ΔT = 150 K → $V = 100\times400\times10^{-6}\times150$ = 6 V.

❖Problem 4 – MHD power

$\sigma = 20$ S/m, $u = 1000$ m/s, $B = 5$ T, channel volume 2 m^3: $P_{max} = \sigma u^2 B^2/4 \times V = 20\times10^6\times25/4\times2 = 250$ MW.

❖Problem 5 – Fuel cell

$\Delta G = -237$ kJ/mol, $n = 2$: $E = 1.23$ V; maximum efficiency = 237/286 = 82.9 %.

14. Important Questions

- Explain OTEC; open and closed cycles with sketches; site selection.
- Tidal power: single basin and double basin schemes; derive power from basin.
- Explain wave energy conversion devices; mini-hydel schemes and economics.
- Need for DEC; limitations of Carnot cycle.
- Seebeck, Peltier, Thomson effects; figure of merit; materials & applications.
- MHD generator: principle, Hall effect, ionisation, open and closed cycles, accelerator.
- Fuel cell: working, Faraday's laws, thermodynamics, fuel selection, types.

All the Best!